

Project Design Phase-II

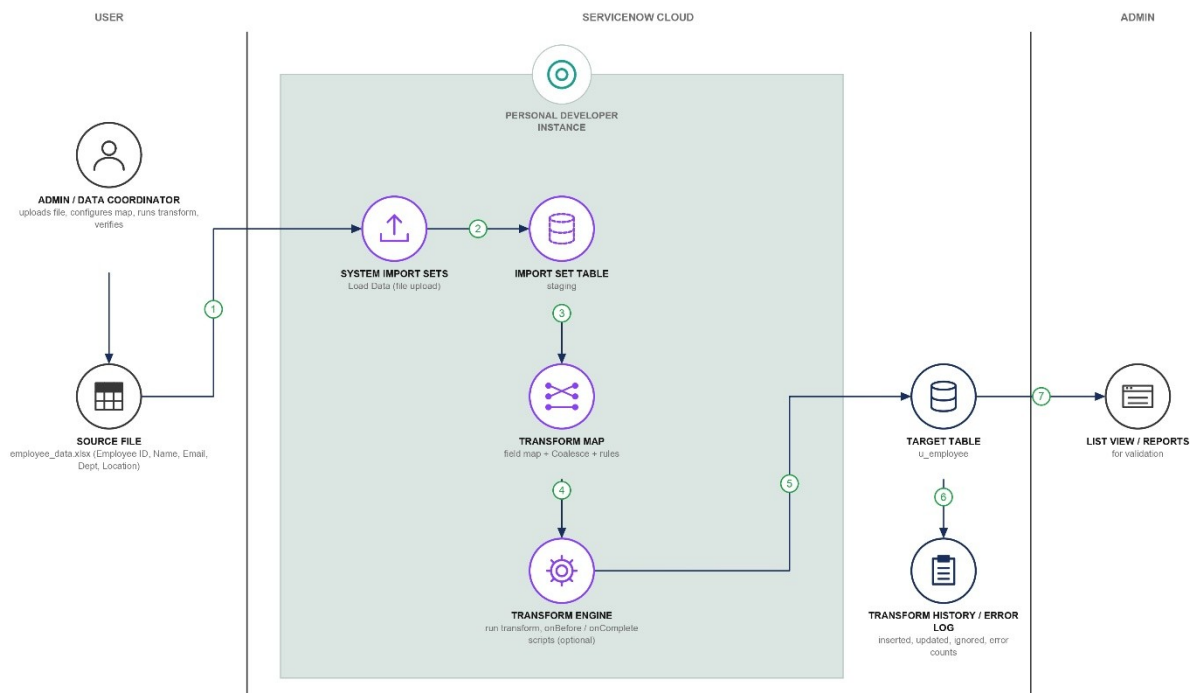
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-1900
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

Technical architecture

Spreadsheet > Import Set > Transform Map > Target Table (ServiceNow Personal Developer Instance)



1. Admin uploads employee_data.xlsx using System Import Sets > Load Data.
2. Rows are loaded into the Import Set Table (staging).
3. Staged rows go to the Transform Map (field map, Coalesce on Employee ID, rules).
4. The Transform Engine runs the transform (onBefore / onComplete scripts optional).
5. The engine inserts / updates records in the Target Table (u_employee).
6. Inserted, updated, ignored and error counts are written to Transform History / Error Log.
7. Admin verifies the data in List View / Reports.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the administrator interacts with the solution	ServiceNow Platform UI (Next Experience / Classic list & form views)
2	Application Logic-1	Loading the spreadsheet into a staging table	System Import Sets > Load Data (Import Set)
3	Application Logic-2	Mapping source fields to target fields and preventing duplicates	Transform Map, Field Maps, Coalesce
4	Application Logic-3	Optional cleaning / validation logic during transform	Transform Map scripts (onBefore / onAfter, JavaScript)
5	Database	Import Set staging table and target table	ServiceNow Now Platform database – u_import_employee, u_employee
6	Cloud Database	Database service on cloud	ServiceNow cloud instance (Personal Developer Instance)
7	File Storage	Source spreadsheet file	Excel (.xlsx) / CSV file; stored as an attachment during import
8	Infrastructure (Server / Cloud)	Application deployment	ServiceNow cloud (PDI from developer.servicenow.com)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source framework is required; source data is prepared in Excel / CSV	Microsoft Excel / CSV
2	Security Implementations	Role-based access to Import Sets and target table; ACLs; encrypted (HTTPS) access to the instance	ServiceNow roles (admin / import_admin), ACLs, HTTPS
3	Scalable Architecture	Same Transform Map can be reused for larger files; platform is multi-tier and cloud hosted	ServiceNow Now Platform
4	Availability	Cloud-hosted instance accessible through a browser	ServiceNow cloud
5	Performance	Bulk load through staging table and batch transform; performance measured by import and transform time	Import Set & Transform engine